	Faculty of Computer Science and Information System **Introduction to Database Systems**	(Midterm) 2025-2026 Time : 1 hour
	Student Name:	

Question 1:

What is the difference between a database schema and a database state?

What four main types of actions involve databases?

Discuss the main characteristics of the database approach and how it differs from traditional file systems.

What are the different types of database end users? Discuss the main activities of each.

Discuss the capabilities that should be provided by a DBMS.

Discuss the main categories of data models. What are the basic differences between the relational model, the object model, and the XML model?

Which of the following is defined as a collection of programs that enables users to create and maintain a database?

- A) Database System
- B) Database Management System (DBMS)
- C) Data Dictionary
- D) Database Catalog

Which of the following commands would you use to find the total sales for each city, but only include cities that have more than $10,000 in total sales?

- A) `WHERE SUM(Sales) > 10000`
- B) `HAVING SUM(Sales) > 10000`
- C) `GROUP BY Sales > 10000`
- D) `WHERE Sales > 10000 GROUP BY City`

. What is the result of a `LEFT JOIN` if there is no matching record in the right table?

- A) The row is excluded from the results.
- B) The row is included, with NULL values for all columns of the right table.
- C) An error is returned by the database.
- D) The row is included, but with zeros (0) for all columns of the right table.

in a `GROUP BY` query, which columns can be included in the `SELECT` statement?

- A) Any column from the table.
- B) Only columns included in the `GROUP BY` clause or used within aggregate functions.
- C) Only the Primary Key.
- D) Only numeric columns.

Which DBMS feature ensures that multiple users can access the database at the same time without conflicts?

- A) Backup and Recovery
- B) Concurrency Control
- C) Data Integrity
- D) Program-Operation Independence

"Workers Behind the Scene" refers to people who:

- A) Use the database for their daily jobs
- B) Design the database schema
- C) Design and implement the DBMS software itself
- D) Authorize access to the database

The information stored in the catalog, which describes the structure and constraints of the database, is called:

- A) Program-Data Independence
- B) Transactional Data
- C) Meta-data
- D) Canned Transaction